

MOHAMMAD AMANOUR RAHMAN

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SUMMARY

Researcher specializing in trustworthy medical image analysis, with a sustained, self-directed research program spanning segmentation, explainability, and uncertainty-aware deep learning, resulting in multiple peer-reviewed journal publications and conference papers as sole or lead author. Motivated by building clinically reliable, interpretable AI systems, with a consistent emphasis on rigorous, honest experimental reporting over inflated claims. Seeking to extend this independent research trajectory into a PhD under close academic mentorship.

RESEARCH INTERESTS

Medical Image Analysis · Deep Learning · Explainable AI (XAI) · Federated Learning · Uncertainty Quantification & Conformal Prediction · Computer Vision · AI in Healthcare · Trustworthy & Multimodal Learning

EDUCATION

Ahsanullah University of Science and Technology (AUST)

B.Sc. in Computer Science and Engineering

2022 – 2026

- **Relevant Coursework:** Artificial Intelligence, Machine Learning, Deep Learning, Pattern Recognition, Digital Image Processing, Soft Computing, Mathematical Analysis for Computer Science.

PUBLICATIONS

Journal Articles

Published

- [1] Rahman, M.A., Prerona, R.Z. "Spectral-Spatial Mamba with Uncertainty-Guided Refinement for Thyroid Nodule Diagnosis." **Array** (Elsevier), 2026. DOI: [10.1016/j.array.2026.101058](#)
- [2] Rahman, M.A. "HyFormer-Net: A Synergistic CNN-Transformer with Interpretable Multi-Scale Fusion for Breast Lesion Segmentation and Classification in Ultrasound Images." **Intelligence-Based Medicine** (Elsevier), 2026. DOI: [10.1016/j.ibmed.2026.100413](#)
- [3] Rahman, M.A. "TCB-Net: Topology-Constrained Boundary-Interior Decoupled Network for Breast Ultrasound Lesion Segmentation." **IET Image Processing**, Accepted for Publication, 2026. Article ID: IPR270445. DOI: [10.1049/ipr2.70445](#)

Under Review

- [4] Rahman, M.A. "Joint Thyroid Nodule Segmentation and TI-RADS-Aligned Boundary Morphology Typing via a Dual-Head Network Trained on Automatically Generated Pixel-Level Labels." **Scientific Reports**. Under Review.
- [5] Rahman, M.A. "OADCAI: Ordinal-Aware Multimodal Deep Learning for Three-Class Pulmonary Nodule Classification Including Indeterminate Cases." **PLOS ONE**. Under Review.

Conference & Workshop Papers

- [6] Rahman, M.A. "MedSaab-US: A Backpropagation-Free Multi-Scale Wavelet-Saab Framework for Thyroid Nodule Segmentation in Ultrasound Images." **IEEE ICIP 2026 Satellite Workshop**. Accepted. arXiv: [10.48550/arXiv.2607.02209](#)
- [7] Rahman, M.A. "RadiomicNet: A Hybrid Radiomics-Guided Lightweight Architecture for Interpretable Medical Image Segmentation." **IEEE ICIP 2026 Satellite Workshop**. Accepted. arXiv: [10.48550/arXiv.2607.02185](#)
- [8] Rahman, M.A. "UCBound-Net: Uncertainty-Guided Boundary-Aware Continual Learning for Domain-Incremental Ultrasound Segmentation." **MICCAI 2026 Workshop on Continual Learning in Medical Multimodal-Vision (CLiMeM 2026)**. Accepted.

RESEARCH EXPERIENCE

Independent Researcher — Medical Image Analysis & Deep Learning

2024 – Present

- Designed and implemented multiple CNN, Transformer, and hybrid deep learning architectures for medical image analysis, as sole or lead author.
- Conducted experiments across breast ultrasound, thyroid ultrasound, CT, and MRI datasets, spanning segmentation, classification, explainability, uncertainty quantification, and federated learning.

- Published 3 journal articles (3 published) and 2 IEEE ICIP workshop papers (accepted); 2 further journal manuscripts currently under review.

Lead Researcher, Undergraduate Thesis

2025 – 2026

- *Automated Brain Hemorrhage Detection, Segmentation, and Severity Quantification from Non-Contrast CT Using a Hybrid Deep Learning Framework.*
- Leading the design and clinical-facing validation of the framework, introducing a Multi-Dimensional Imaging Severity Index (MDISI), with the goal of collaboration with a neurosurgeon for clinical validation; intended for submission to a leading medical image analysis journal.
- Presented the thesis research as a poster presentation at AUST.

ACADEMIC SERVICE

Peer Reviewer

- PLOS ONE — 1 manuscript
- Efficient Medical AI (EMA4MICCAI) Workshop, MICCAI 2026 — 3 manuscripts
- NED University Journal of Research — 2 manuscripts
- Jordanian Journal of Computers and Information Technology — 1 manuscript

Open-Source Contributor — Project MONAI (NVIDIA & King's College London)

- **Implemented 5-Fold Cross-Validation for UNETR** (PR #422, MONAI Research-Contributions): added a `--fold` argument and integrated MONAI's `partition_dataset` API for fold-indexed training/validation.
- **Resolved High-Version MONAI Compatibility Issues** (PR #421): fixed a crash from the deprecated `pos_embed` parameter in ViT under MONAI 1.3.0+, using a version-aware runtime compatibility mechanism built with Python's `inspect` module.
- Resolved a documentation defect in the PerceptualLoss module (Issue #8592), improving usability and reproducibility for the broader community.

RESEARCH SOFTWARE

MedXAI — Open-Source Python Toolkit for Medical Imaging AI

[PyPI](#) | [GitHub](#)

- Independently developed and maintains MedXAI, a lightweight open-source toolkit for reproducible medical imaging AI research, distributed via PyPI and GitHub.
- Provides modular utilities for segmentation metrics, custom loss functions, image preprocessing, radiomics feature extraction, and explainability (Grad-CAM) for PyTorch-based research workflows.

EARLIER PROJECTS

EfficientNet-CBAM for Brain Tumor Classification

- Developed an EfficientNetB3-CBAM framework for multi-class brain tumor MRI classification, achieving 99.29% accuracy; integrated HiResCAM for explainability. Tech: TensorFlow, Keras, OpenCV.

Deforestation Mapping in Dhaka — Semantic Segmentation

- Built a high-resolution satellite imagery dataset (2012–2022) and benchmarked U-Net variants (Dice, IoU, F1, Precision, Recall) for deforestation mapping. Tech: Python, TensorFlow, PyTorch, OpenCV.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, MATLAB

Deep Learning Frameworks: PyTorch, TensorFlow, Keras, MONAI

Computer Vision & Image Processing: OpenCV, scikit-image, Albumentations, Pillow

Scientific Computing & Data Analysis: NumPy, SciPy, Pandas, Matplotlib

Research & Software Development: Git, GitHub, GitHub Actions, PyPI, Jupyter Notebook, Google Colab, VS Code

Documentation: LaTeX, Overleaf